Solution exercises second chapter Modern Cosmology

Abstract

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1 Exercise 1

Convert the following quantities by inserting the appropiate factors of $c, \hbar$ and k_b:

1. $T_0 = 2.276K \rightarrow eV$
2. $\rho_\gamma = \pi^2 \frac{T_0^4}{15} \rightarrow eV^4$ *and* gcm^{-3}
3. $\frac{1}{H_0} \rightarrow cm$
4. $m_{Planck} \equiv \sqrt{\frac{\hbar c}{G}} = 1.2 \cdot 10^9 GeV \rightarrow K, cm^{-1}, s^{-1}$

To convert the above quantities, multiply and divide by the factor which yield, after setting $\hbar$, c or k_b equal to 1, that expression.

$$
\begin{aligned}
T_0 &= T_0 \cdot \frac{k_b}{k_b} = 1.9612 \cdot 10^{-4} \frac{\mathrm{eV}}{k_b} \overset{k_b=1}{=} 1.9612 \cdot 10^{-4} \mathrm{eV} \\
\rho_\gamma &= \pi^2 \frac{T_0^4}{15} = 9.7347 \cdot 10^{-16} \left[\mathrm{eV}^4\right] = 20.0327 (\hbar \cdot c)^3 \left[\frac{\mathrm{eV}^4}{\mathrm{eV}^3 \mathrm{L}^3}\right] \\
&= 7.4724 \cdot 10^{-36} \left[\frac{\mathrm{eV}}{\mathrm{L}^3}\right] \overset{c=1=\hbar}{=} 7.4724 \cdot 10^{-33} \frac{\mathrm{g}}{\mathrm{cm}^{-3}} \\
\frac{1}{H_0} &= 4.4131 \cdot 10^{17} \left[\mathrm{E}^{-1}\right] = 4.4131 \cdot \frac{1}{\hbar c} \cdot 10^{17} \left[\mathrm{L}\right] = 2.2372 \cdot 10^{24} \left[\mathrm{L}\right] \\
&= 2.2372 \cdot 10^{26} \mathrm{cm} \\
m_{\mathrm{Planck}} &= 1.2 \cdot 10^{19} \left[\mathrm{E}\right] = 1.2 \cdot 10^{19} \cdot k_b \left[\frac{\mathrm{E}}{\mathrm{E} \cdot \mathrm{Tem}^{-1}}\right] = 1.0340 \cdot 10^{11} \mathrm{K} \\
&= 1.2 \cdot 10^{19} \hbar c \left[\frac{\mathrm{E}}{\mathrm{E} \cdot \mathrm{L}}\right] = 2.3672 \cdot 10^{10} \mathrm{cm}^{-1} \\
&= 1.2 \cdot 10^{19} \hbar \left[\frac{\mathrm{E}}{\mathrm{E} \cdot \mathrm{T}^{-1}}\right] = 0.7896 \, \mathrm{s}^{-1}
\end{aligned}
$$

2 Exercise 2

Show that the geodesic equation gets the correct equations of motion for a particle travelling freely in two dimensions using polar coordinates.

First of all, the equations of motion for a particle travelling freely in two dimensions, using polar coordinates
$\begin{pmatrix} r & \theta \end{pmatrix}$, is

$$\begin{cases} \ddot{r} = r\dot{\theta}^2 \\ \frac{d}{dt}(mr^2\dot{\theta}) = 0 \end{cases}$$

We follow two ways.
We use the equations

$$\frac{d^2y^l}{dt^2} + \left[\left(\left\{ \frac{\partial x}{\partial y} \right\}^{-1} \right)^l_i \frac{\partial^2 x^i}{\partial y^j \partial y^k} \right] \frac{dy^k}{dt} \frac{dy^j}{dt} = 0$$

Now, we identify

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} r\cos(\theta) \\ r\sin(\theta) \end{pmatrix}$$
$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} r \\ \theta \end{pmatrix}$$

First, the jacobian matrix is

$$\frac{\partial \mathbf{x}}{\partial \mathbf{y}} = \begin{pmatrix} \cos(\theta) & -r\sin(\theta) \\ \sin(\theta) & r\cos(\theta) \end{pmatrix}$$

Its inverse can be computed with various method, yielding

$$A = \left\{ \frac{\partial x}{\partial y} \right\}^{-1} = \begin{pmatrix} \cos(\theta) & \sin(\theta) \\ -\frac{\sin(\theta)}{r} & \frac{\cos(\theta)}{r} \end{pmatrix}$$

(maybe note the fact that the numerical inverse is right in this case since is the inverse of the jacobian and not the contrary, in which case one should evaluate it in the old points $\mathbf{x}$)

Now, for the hessian matrix, this is

$$\frac{\partial r\cos(\theta)}{\mathbf{y}^2} = \begin{pmatrix} 0 & -\sin(\theta) \\ -\sin(\theta) & -r\cos(\theta) \end{pmatrix}$$
$$\frac{\partial r\sin(\theta)}{\mathbf{y}^2} = \begin{pmatrix} 0 & \cos(\theta) \\ \cos(\theta) & -r\sin(\theta) \end{pmatrix}$$

So, by matrix multiplication, we get the two equations of motion. The equation of motion for the radius is

$$\frac{\mathrm{d}^2 r}{dt^2} + \left[A_i^1 \frac{\partial x^i}{\partial y^j \partial y^k}\right] \frac{dy^j}{dt}\frac{dy^k}{dt} =$$

$$= \frac{d^2 r}{dt^2} + \cos(\theta)\left(-2\sin(\theta)\dot{r}\dot{\theta} - rcos(\theta)\dot{\theta}^2\right) + 2\sin(\theta)\left(\cos(\theta)\dot{r}\dot{\theta} - r\sin(\theta)\dot{\theta}\right.$$

$$= \frac{d^2 r}{dt^2} - r\dot{\theta}^2 = 0$$

The equation of motion for the angle is

$$\frac{\mathrm{d}^2 r}{dt^2} + \left[A_i^2 \frac{\partial x^i}{\partial y^j \partial y^k}\right] \frac{dy^j}{dt}\frac{dy^k}{dt} =$$

$$= \frac{\mathrm{d}^2\theta}{\mathrm{d}t} + \left(-2\frac{\sin(\theta)}{r}(-\sin(\theta))\dot{r}\dot{\theta} + \frac{\sin(\theta)}{r} r\cos(\theta)\dot{\theta}^2\right) +$$

$$+ \left(2\frac{\cos(\theta)}{r}\cos(\theta)\dot{r}\dot{\theta} + \frac{\cos(\theta)}{r}(-r\sin(\theta))\dot{\theta}^2\right)$$

$$= \frac{\mathrm{d}^2\theta}{\mathrm{d}t^2} + 2\frac{1}{r}\dot{r}\dot{\theta} = 0$$

Assuming the mass is time independent, we can rewrite the last line as

$$\frac{\mathrm{d}}{\mathrm{d}t}(mr^2\dot{\theta}) = 0$$

Now we turn to the second way. here we use the fact that the manifold we are considering is torsionless, so we may write the Christoffel symbol depending only on the metric. The metric induce by the spherical coordinates is

$$\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\,\mathrm{d}\theta^2$$

From this we read off the metric coefficients,

$$(g_{\alpha\beta}) = \begin{pmatrix} 1 & \\ & r^2 \end{pmatrix}$$

Now we begin to compute the Christoffel Symbols.

$$\Gamma^1_{\beta\gamma} = \frac{1}{2}g^{1\rho}(g_{1\gamma,\beta} + g_{1\gamma,\beta} - g_{\beta\gamma,1}) \overset{\beta=2=\gamma}{=} -\frac{1}{2}\partial_r r^2 = -r$$

$$\Gamma^2_{\beta\gamma} = \frac{1}{2}g^{2\rho}(g_{2\beta,\gamma} + g_{2\gamma,\beta} - \cancelto{0}{\beta\gamma,\rho}) = \frac{1}{2}r^{-2}\partial_r r^2 = r^{-1}$$

We turn to the geodesic equation to find the motion of the particle in the plane,

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - \Gamma^1_{22}\left(\frac{\mathrm{d}y^2}{\mathrm{d}t}\right)^2 = \frac{\mathrm{d}^2 r}{\mathrm{d}t^2} - r\dot{\theta}^2 = 0$$

$$\frac{\mathrm{d}^2\theta}{\mathrm{d}t} + \Gamma^1_{12}\dot{r}\dot{\theta} + \Gamma^1_{21}\dot{\theta}\dot{r} = \frac{\mathrm{d}^2\theta}{\mathrm{d}t} + 2\frac{1}{r}\dot{r}\dot{\theta} = 0$$

which we can recast in the same form as the one given in the preamble.

3 Exercise 3

Find how the energy of a massive, nonrelativistic particle changes as the universe expands, i.e a FLRW universe.

First we need to know how the particle travels in a FLRW universe. This can be achieved by looking at the geodesic equations, with Christoffel Symbols

$$\frac{\mathrm{d}^2 x^\alpha}{\mathrm{d}\lambda} + \Gamma^\alpha_{\beta\gamma} \frac{\mathrm{d}x^\beta}{\mathrm{d}\lambda} \frac{\mathrm{d}x^\gamma}{\mathrm{d}\lambda} = 0$$

Notice that, contrary to exercise 2, we have an affine parameter (λ) which may not coincide with time[1]. Let's define the momentum of this particle, which is

$$\mathbf{P} = \left(E, \vec{p}\right) \overset{!}{\equiv} \left(\frac{\mathrm{d}x^\alpha}{\mathrm{d}\lambda}\right)$$

With this, we can rewrite the geodesic equation as

$$\frac{\mathrm{d}P^\alpha}{\mathrm{d}\lambda} + \Gamma^\alpha_{\beta\gamma} P^\beta P^\gamma = 0$$

Notice that, although we have a general affine parameter, we may as well express its derivatives with respect to time. That is, by the chain rule,

$$\frac{\mathrm{d}}{\mathrm{d}\lambda} = \overbrace{\frac{\mathrm{d}x^0}{\mathrm{d}\lambda}}^{E} \frac{\mathrm{d}}{\mathrm{d}t} = E \frac{\mathrm{d}}{\mathrm{d}t}$$

Finally,

$$E \frac{\mathrm{d}P^\alpha}{\mathrm{d}t} + \Gamma^\alpha_{\beta\gamma} P^\beta P^\gamma = 0$$

We are asked for the time behaviour of the particle in motion, therefore let's concentrate on the 0th component of P^α. With this, the equations of motion tell us

$$E \cdot \dot{E} = -\delta_{ij} \dot{a} a P^i P_j = \frac{m^2 - E^2}{a} \dot{a}$$

Putting all together, we find that the energy is

$$\frac{\dot{E}}{E} + \frac{\dot{a}}{a}\left(1 - \frac{m^2}{E^2}\right) = 0$$

[1] The beauty of General relativity if you ask me.

4 Exercise 4

Show that the geodesic equation in an euclidean universe implies $\frac{\mathrm{d}^2\vec{x}}{\mathrm{d}\eta^2} = 0$*, where* η *is the conformal time. This, in turn, implies that in comoving coordinates* $\left(\eta, \boldsymbol{x}\right)$*, photons travel on straight lines, as in Minkowsky space.*

The metric of an euclidean universe is

$$(g_{\mu\nu}) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

We can compute the Christoffel symbols for this metric, which are $\Gamma^{\alpha}_{\beta\gamma} = 0$ for any choice of $\begin{pmatrix} \alpha & \beta & \gamma \end{pmatrix}$. Therefore, the geodesic equations say that

$$\frac{\mathrm{d}^2\mathbf{x}}{\mathrm{d}\lambda^2} = \mathbf{0}_{\mathbb{R}^4}$$

The objective is now to convert that arbitrary affine parameter λ in η. We employ the chain rule to change dependences in the derivatives, that is

$$\begin{aligned} \frac{\mathrm{d}^2 x^\alpha}{\mathrm{d}\lambda^2} &= \frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{\mathrm{d}x^\alpha}{\mathrm{d}\eta}\frac{\mathrm{d}\eta}{\mathrm{d}\lambda}\right) = \frac{\mathrm{d}\eta}{\mathrm{d}\lambda}\frac{\mathrm{d}}{\mathrm{d}\eta}\left(\frac{\mathrm{d}x^\alpha}{\mathrm{d}\eta}\frac{\mathrm{d}\eta}{\mathrm{d}\lambda}\right) \\ &= \frac{\mathrm{d}^2 x^\alpha}{\mathrm{d}\eta^2}\left(\frac{\mathrm{d}\eta}{\mathrm{d}\lambda}\right)^2 + \frac{\mathrm{d}x^\alpha}{\mathrm{d}\eta^2}\cdot\frac{\mathrm{d}^2\eta}{\mathrm{d}\lambda\,\mathrm{d}\eta} = 0 \end{aligned}$$

Since the term $\frac{\mathrm{d}^2\eta}{\mathrm{d}\lambda\,\mathrm{d}\eta} = 0$, then we have the relationship connecting every component with the conformal time, that is

$$\frac{\mathrm{d}^2 x^\alpha}{\mathrm{d}\lambda^2} = \frac{\mathrm{d}^2 x^\alpha}{\mathrm{d}\eta^2}\left(\frac{\mathrm{d}\eta}{\mathrm{d}\lambda}\right)^2$$

Therefore, in an euclidean universe, we indeed have that

$$\frac{\mathrm{d}^2\vec{x}}{\mathrm{d}\eta^2} = 0$$

5 Exercise 5

At early times, the cosmological constant can be neglected. Using this approximation, integrate $H^2(t) = \frac{8\pi G}{3}\left[\rho(t) + \frac{\rho_{cr}-\rho(t_0)}{a^2(t)}\right]$ *to obtain* $a(t)$*. In turn, using* $T(t) = \frac{T_0}{a(t)}$*, determine the times when the cosmic temperature was* $0.1MeV$ *and* $\frac{1}{4}eV$*.*[2]

To solve this, let's suppose that the equation of state at these early times was time independent, that is we have $\rho = \rho_0 a^{-3(1+\omega)}$. Now we massage the Friedmann equation,

$$\frac{\dot{a}}{a} = \sqrt{\frac{8\pi G}{3}\left[\rho_0 a^{-3(1+\omega)} + \frac{\rho_{\text{cr}} - \rho_0}{a^2(t)}\right]} = \frac{1}{a}\sqrt{\frac{8\pi G}{3}\left[\rho_0 a^{2-3(1+\omega)} + \rho_{\text{cr}} + \rho_0\right]}$$
$$= \frac{1}{a}\sqrt{\frac{8\pi G}{3}}\sqrt{1 + \frac{\rho_0}{\rho_0 + \rho_{\text{cr}}} a^{2-3(1+\omega)}}$$

Therefore,

$$\int_{a'}^{1} \frac{1}{\sqrt{1 + \frac{\rho_0}{\rho_0+\rho_{\text{cr}}} a^{2-3(1+\omega)}}}\, \mathrm{d}a = \int_{t(a)}^{0} 1\ \mathrm{d}t = -\sqrt{\frac{8\pi G}{3}} t$$

For the left hand side, use the substitution $y = \left(\frac{\rho_0}{\rho_0+\rho_{\text{cr}}}\right)^{\frac{1}{2-3(1+\omega)}} a$, so that

$$\left(\frac{\rho_0}{\rho_0 + \rho_{\text{cr}}}\right)^{-\frac{1}{2-3(1+\omega)}} \int_{y'\left(\frac{\rho_0}{\rho_0+\rho_{\text{cr}}}\right)^{\frac{1}{2-3(1+\omega)}}}^{\left(\frac{\rho_0}{\rho_0+\rho_{\text{cr}}}\right)^{\frac{1}{2-3(1+\omega)}}} \frac{1}{\sqrt{1 + y^{2-3(1+\omega)}}}\, \mathrm{d}y = -\sqrt{\frac{8\pi G}{3}} t$$

Let's suppose that we are deep into the radiation era, so that $\omega = \frac{1}{3}$. The integral simplifies a lot:

$$\int \frac{1}{\sqrt{1 + y^{-2}}}\, dy = \sqrt{1 + y^2} + C$$

Furthemore, experimental evidence show that ρ_{cr} is within the error bars of the measured energy density ρ_0, therefore we may as well suppose that $\rho_0 \sim \rho_{\text{cr}}$. After all these assumptions, the equation simplifies to

$$\sqrt{1 + y^2}\Big|_{y'\sqrt{2}}^{\sqrt{2}} = -\sqrt{\frac{16\pi G}{3}} t$$
$$\sqrt{3} - \sqrt{1 + 2(y)^2} = -\sqrt{\frac{16\pi G}{3}} t$$

Now, we substitute y in terms of a, to get

[2] This temperature coincide with Big Bang nucleosynthesis and recombination

$$a_{\text{rad}} = \frac{1}{2}\sqrt{\left(\sqrt{3} + \sqrt{\frac{16\pi G}{3}}t\right)^2 - 1}$$

Before proceeding, we must know this to be but an assumption, an estimate of what we believe to be the extense of the radiation era. Precisely, in the FLRW universe we may as well suppose that the temperature 0.1MeV and $\frac{1}{4}$eV are indeed well into the radiation epoch. We only get a matter dominated FLRW universe at roughly 10^{-6}MeV.

6 Exercise 6

Derive some expressions for the conformal time η as a function a.

1. *Show that $\eta \propto a^{\frac{1}{2}}$ in a matter dominated universe, while $\eta \propto a$ in one dominated by radiation.*

2. *Consider a Euclidean concordance universe with equality at a_{eq}, at early times so that the effect of Λ can be neglected. Show that*

$$\eta = \frac{2}{\sqrt{\Omega_{\mathrm{m}}} H_0^2} \left[a + a_{\mathrm{eq}} - \sqrt{a_{\mathrm{eq}}} \right]$$

   What is the conformal time at $z = 1100$?

To answer the first part, we recall that in a matter dominated universe $a \propto t^{\frac{2}{3}}$, while in a radiation dominated universe $a \propto t^{\frac{1}{2}}$.

7 Exercise 7

Consider a galaxy of physical sixe $5kpc$*. What angle would this galaxy subtend if situated at redshift* 0.1*? What about redshift* 1*? Do the calcualtion in a Euclidean universe, first matter dominated and then for the fiducial concordance cosmology.*

8 Exercise 8

How is the energy density of a gas of photons with a black-body spectrum related to the specific intensity of the radiation? That is, what is the relation between ρ_γ and I_ν?

9 Exercise 9

1. *Compute the pressure of a relativistic species in equilibrium with temperature T. Show that $\mathcal{P} \stackrel{!}{=} \frac{\rho}{3}$ for both the Fermi-Dirac and Bose-Einstein statistics.*

2. *Suppose the distribution function depends only on $\frac{E}{T}$ as it does in equilibrium in the absence of chemical potential. Find $\frac{d\mathcal{P}}{dt}$.*

10 Exercise 10

Plot $d_L, d_A(z)$ and $m-M$ as a function of redshift for a Euclidean, matter dominated universe (this can be done analytically) and for the fiducial Euclidean concordance cosmology (for this, evaluate numerically a $1D$ integral). Neglect K correction for $m-M$.

11 Exercise 11

Consider the entropy density $s \equiv \frac{\rho+\mathcal{P}}{T}$. *For a massless particle, we showed back in exercise* 9 *that* $\mathcal{P} = \frac{\rho}{3}$ *so* $s = \frac{4\rho}{3T}$. *Express* s *as a function of* T *for both bosons and fermions (assumed massless) in equilibrium with zero chemical potential. Show that the entropy density for a massive particle in equilibrium, i.e where bath temperature* $T \ll m$ *with* $\mu = 0$, *is exponentally suppressed.*

12 Exercise 12

Show that the number density of one generation of neutrinos and anti-neutrinos in the universe today is

$$n_{\nu_i} = \frac{3}{11} n_\gamma = 112\text{cm}^{-3}$$

13 Exercise 13

consider the following two scenarios. Each has energy density equal to the critical density divided up between only two components: a cold, dark matter particle and a neutrino. The neutrino in each case has the standard abundance and temperature. The only difference between the two secnarios is in one the neutrino is massless, while in the other it has a mass of $0.06eV$*. Plot the energy density as a function of scale factor in each of these scenarios.*

14 Exercise 14

Determine the epoch of dark energy-matter equality assuming that the dark energy is a cosmological constant.